SAMSUNG

Big Data Course

Capstone Project Action Plan

For students (instructor's review required)

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Course	Big Data Course
Team Name	Team 6
Team Leader/ Members	Vàng A Ký / Nguyễn Đình Bằng, Vàng Thị Dẩm, Dương Thị Hạnh, Ngô Lương Nguyên
Project Title	SentimentGuard: Real-time Sentiment Classification of Live Chat Comments for Early Detection of Harmful Comment Spikes

Goal

Build a real-time pipeline that classifies live chat comments as negative, neutral or positive using Apache Kafka and Spark Structured Streaming.
Detect a spike of negative comments within a short time window and alert the streamer while the session is still running.
Train and evaluate a Spark ML text classifier, reporting macro-F1 and per-class precision, recall and F1.
Deliver a low-cost, self-hosted prototype that a streamer with no moderation team can run on commodity hardware.
State clearly which results come from replayed data and which guarantees remain unverified.

Abstract

Live chat moves far faster than a solo streamer can read, so harmful comment spikes are usually noticed only after the damage is done. This project builds a real-time sentiment classification pipeline: comments are ingested through Apache Kafka, processed by Spark Structured Streaming with event-time windows and watermarks, and scored by a Spark ML model into negative, neutral and positive. Results are aggregated per one-minute window and an alert is raised when the negative ratio crosses a configurable threshold. Because labelled live chat is not publicly available and platform terms restrict retaining user comments, the model is trained on a public, CC0-licensed IMDb review dataset, with the three sentiment classes derived from user ratings, and the live stream is reproduced by replaying that data through Kafka.

Method

The project follows a five-stage pipeline: Kafka ingestion, Spark Structured Streaming transformation, Spark ML classification, Spark SQL query and insight, and Streamlit visualization. Text cleaning is implemented once in a shared module and reused by both the training and the streaming code, so the two cannot drift apart. The classifier is a

Spark ML Pipeline chaining tokenization, stop-word removal, n-grams and TF-IDF with a linear model, with TF-IDF fitted only on the training split to prevent leakage. Duplicate and near-duplicate comments are removed before the train/validation/test split, the test set is used once at the end, and class imbalance is handled with class weights.

Data

The model is trained on a public IMDb review dataset under CC0 (about 1.0 million reviews; source and licence in data/raw/SOURCE.md). It carries no sentiment labels: the three classes are derived from the 1-10 user ratings by a documented mapping, reported as rating-derived labels. Live chat is not available with sentiment labels and platform terms restrict retaining it, so the stream is simulated: the dataset is replayed through Kafka at a configurable rate, with bursts and deliberately late records to exercise watermarking and the dead-letter queue. Labels never enter the stream; they are stored separately and joined back by comment ID for evaluation. User identifiers are replaced with salted hashes and no personal data is retained.

Expected Outcome

The prototype is expected to classify comments in near real time and to raise an alert within seconds of a harmful comment spike, with end-to-end latency reported at p50, p95 and p99. Model quality will be reported as macro-F1 with per-class precision, recall and F1, alongside measured throughput and the results of restarting the job mid-stream. The benefit is for people who stream alone and cannot afford commercial moderation tools: an early warning lets them slow the chat, enable moderation mode or step away before a spike escalates. Because the tool is self-hosted, no comment data leaves the operator's own machine.

Role by Member

Vàng A Ký (Team Leader) - project management, schedule and WBS, report and slide coordination, integration testing.
Nguyễn Đình Bằng - Kafka data engineer: broker setup, topics, replay producer, ingestion throughput measurement.
Vàng Thị Dẳm - streaming engineer: Structured Streaming job, event-time windows, watermark, dead-letter queue, fault tolerance.
Dương Thị Hạnh - machine learning engineer: data preparation, Spark ML pipeline, model training and evaluation.
Ngô Lương Nguyên - data analyst: Spark SQL queries, visualization, Streamlit dashboard, insights.

Schedule Summary	

Phase 1 (Day 1-5) - Planning: team building, action plan, WBS, topic framing and data selection, preliminary presentation.
Phase 2 (Day 6-11) - Prototype: Kafka and Spark setup, ingestion, training data preparation, Spark ML model, Structured Streaming job, queries and dashboard, prototype presentation.
Phase 3 (Day 12-20) - Refinement and delivery: benchmarking, fault-tolerance testing, model tuning, usage scenario, final report and final presentation.
The detailed day-by-day plan is maintained in the accompanying WBS workbook, reviewed on Day 9, Day 14 and Day 19.

Comment & Assessment	